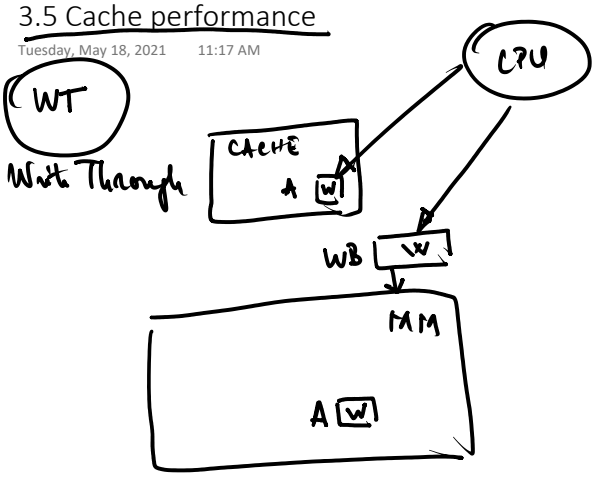
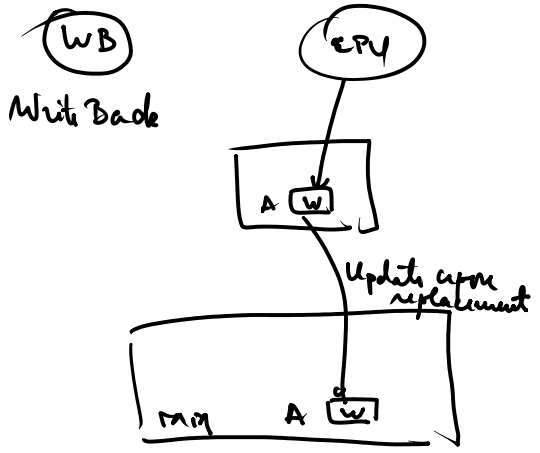
3.5 Cache performance

Tuesday, May 18, 2021 11:17 AM





Parameters used

- Miss Rate

The 3 Sources of Misses

- **Compulsory** – cold start misses
  - fixed by bigger block sizes
- **Conflict** – multiple MM blocks are mapped at the same cache index
  - fixed by increasing SA
  - 32, 64-way SA ≃ FA
- **Capacity**

- % CPU time required by cache imperfections

Hit Rate + Miss Rate = 1

$$\text{CPU}_{time} = IC \times \underbrace{CPI}_{\text{perfect cache}} \times \text{Clock cycle time}$$

- Average Memory Access Time = AMAT ; t_{ac} – access time in the cache

$$AMAT = \text{Hit Rate} \times t_{ac} + \text{Miss Rate} \times (t_{ac} + \underbrace{\text{Miss Penalty}}_{\text{Time Miss Penalty}}) =$$

$$= t_{ac} \times (\text{Hit Rate} + \text{Miss Rate}) + \text{Miss Rate} \times \text{Miss Penalty} =$$

$$= \boxed{t_{ac} + \text{Miss Rate} \times \text{Miss Penalty} = AMAT}$$

- **CPU time**

$$\text{CPU}_{time} = (\text{CPU execution clock cycles} + \underbrace{\text{Memory stall clock cycles}}_{=0 \text{ for a cache hit}}) \times \text{Clock cycle time}$$

CPU execution clock cycles $= IC \times CPI_{ideal}$

Memory stall clock cycles $= \frac{\text{Reads}}{\text{Program}} \times \text{Read miss rate} \times \text{Read penalty} +$

$\frac{\text{Writes}}{\text{Program}} \times \text{Write miss rate} \times \text{Write penalty}$

$= IC \times \left(\frac{\text{Reads}}{\text{Instruction}} \times \text{Read miss rate} \times \text{Read penalty} + \right.$

$\left. + \frac{\text{Writes}}{\text{Instruction}} \times \text{Write miss rate} \times \text{Write penalty} \right)$

$$CPU_{time} = IC \times \left(\underset{(\text{execution})}{CPI_{ideal}} + \frac{\text{Reads}}{\text{Instruction}} \times \text{Read miss rate} \times \text{Read penalty} + \right.$$

$$\left. + \frac{\text{Writes}}{\text{Instruction}} \times \text{Write miss rate} \times \text{Write penalty} \right) \times \text{Clock cycle time}$$

$$CPU_{time} = IC \times \left(CPI_{ideal} + \overbrace{\frac{\text{Memory accesses}}{\text{Instruction}} \times \text{Miss Rate}}^{\text{Misses Per Instruction}} \times \text{Miss Penalty} \right) \times$$

$$\times \text{Clock cycle time}$$

Examples

- CPI = 8.5 c.c.
- Memory accesses per instruction = 3
- Miss Rate = 11%
- Miss penalty = 6 c.c.
- tac = 5 ns = Clock cycle time

$$AMAT = 5\,ns + 0.11 \times \underbrace{6\,c.c. \times 5\,ns}_{\text{Time miss penalty}} =$$

$$= \boxed{8.3\,ns}$$

Time overhead brought by cache imperfections

$$= \boxed{3.3\,ns}$$

$$CPU_{time} = IC \times \left(8.5\,c.c + \underbrace{3 \times 0.11}\times 6\,c.c \right) \times 5\,ns$$

$$CPU_{time\,ideal} = IC \times 8.5\,c.c \times 5\,ns$$

$$\frac{CPU_{time}}{} = \frac{\cancel{IC} \times (1.98\,cc + 8.5\,c.c) \times \cancel{5\,ns}}{} = 1.23$$

$$\frac{\text{CPU time}}{\text{CPU time ideal}} = \frac{\cancel{IC} \times (1.98\ c.c + 8.5\ c.c) \times \cancel{5\ ns}}{\cancel{IC} \times 8.5\ c.c. \times \cancel{5\ ns}} = 1.23$$

=> Because of the cache imperfection => 23% increase of CPU time

Example 2

64KiB cache

Clock cycle time = 20 ns

$\frac{\text{Memory accesses}}{\text{Instruction}} = 1.3$

CPI ideal = 1.5 c.c.

For S.A. solutions => additional MUX on the datapath => clock frequency degradation / decrease of 8.5%

- 1-way SA Vs. 2-way SA ?

statistical measurements

Miss rate$_{1\text{-way}}$ = 3.9%

Miss rate$_{2\text{-way}}$ = 3%

AMAT ✓

CPU time ✓

Miss penalty = 200 ns

AMAT$_{1\text{-way}}$ = 20 ns + 0.039 × 200 ns = 27.8 ns

AMAT$_{2\text{-way}}$ = 21.7 ns + 0.03 × 200 ns = 27.7 ns

Better 2-way

Clock cycle time$_{1\text{-way}}$ = 20 ns

Clock cycle time$_{2\text{-way}}$ = 20 ns × 1.085 = 21.7 ns

CPU time$_{1\text{-way}}$ = IC × (1.5 + 1.3 × 0.039 × Miss penalty in c.c.) × Clock cycle time

= IC × (1.5 × Clock cycle time + 1.3 × 0.039 × Miss penalty in ns)